

Highlights

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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- SC-FMA converts intermediate knowledge artifacts into structured audit records for fixed-budget curation.
- Graph-aware representation fields record fidelity, structural dependency, redundancy, bottleneck exposure, and maintenance actions.
- The PRM800K process-annotation dataset is used to analyze annotation-order preservation as representation behavior under an audit budget.
- A Countries-KG diagnostic graph shows how typed relations alter structural-dependency allocation.
- Audit-record construction supports fixed-budget knowledge maintenance, curation, and reuse.